



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.2

Volume with Whole Number Edges

Flagship

Lesson 10-1-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

CLASS LEGACY MISSION

Seal the Time Capsule

You are the lead builder for the school's 50-year time capsule project. Every memory must fit inside rectangular containers, and the principal needs to know exactly how much each one holds before they are buried. Master volume and the capsule gets sealed for the future.



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism with whole-number edges using $\text{length} \times \text{width} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Length, width, height

Net

Edge



Tap any word to see what it means and a picture.

1

The amount of space inside a three-dimensional solid is its

2

A solid with six rectangular faces, like a box, is a

3

The unit used to measure volume, like cubic centimeters, is

4

The three edge measurements of a rectangular prism are its

5

A flat pattern that folds up into a solid is a

6

A line segment where two faces of a solid meet is an

Reflect — Exit Ticket

A rectangular prism has $l = 11$ in, $w = 5$ in, $h = 4$ in. What is the volume?

- A. 220 in^3
- B. 55 in^3
- C. 220 in^2
- D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Volume with Whole Number Edges is skipping the key idea: "To find the volume of a box, multiply all three edges: $V = \text{length} \times \text{width} \times \text{height}$." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 3 in — $V = l \times w \times h \rightarrow 72 = 6 \times 4 \times h \rightarrow 72 = 24 \times h \rightarrow h = 72 \div 24 = 3 \text{ in.}$
9. **Try It 2:** A. 200 cubic inches — $V = l \times w \times h = 10 \times 4 \times 5 = 200 \text{ cubic inches.}$ *Volume always uses cubic units because it fills space in three directions.*
10. **Exit Ticket:** A. 220 in^3 — $V = 11 \times 5 \times 4 = 220 \text{ cubic inches.}$ *Remember: volume uses cubic units (in^3).*